


# Topics to be covered

Revised ~~3~~ 8 1/2

- **A** COM of Continuous System
- **B** Combinational Problems
- **C** Questions
- **D**



**QUESTION 1** HW

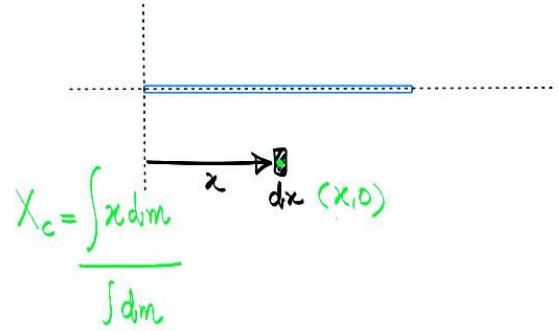
A rod of length $L$ has non-uniform linear mass density given by $\lambda(x) = a + b\left(\frac{x}{L}\right)^2$, where $a$ and $b$ are constants and $0 \le x \le L$. The value of location of centre of mass of the rod is at :

**A** $\frac{4}{3}\left(\frac{a+b}{2a+3b}\right)L$

**B** $\frac{3}{4}\left(\frac{2a+b}{3a+b}\right)L$ Ans.

**C** $\frac{3}{2}\left(\frac{2a+b}{3a+b}\right)L$

**D** $\frac{3}{2}\left(\frac{a+b}{2a+b}\right)L$




$X_c = \frac{\int x\,dm}{\int dm}$

$\lambda = a + \frac{bx^2}{L^2}$

$dm = \lambda\,dx$

$dm = \left(a + \frac{bx^2}{L^2}\right)dx$

$$X_c = \frac{\int_0^L x\left(a+\frac{bx^2}{L^2}\right)dx}{\int_0^L \left(a+\frac{bx^2}{L^2}\right)dx} \quad \text{Variable.}$$

$$= \frac{\int_0^L ax\,dx + \int_0^L \frac{bx^3}{L^2}dx}{\int_0^L a\,dx + \int_0^L \frac{bx^2}{L^2}dx}$$

$$= \frac{a\left[\frac{x^2}{2}\right]_0^L + \frac{b}{L^2}\left[\frac{x^4}{4}\right]_0^L}{a[x]_0^L + \frac{b}{L^2}\left[\frac{x^3}{3}\right]_0^L} = \frac{\frac{aL^2}{2} + \frac{bL^{\cancel{4}2}}{4\cancel{L^2}}}{aL + \frac{bL^{\cancel{3}2}}{3\cancel{L}}}$$

$$= L\left[\frac{\frac{a}{2}+\frac{b}{4}}{a+\frac{b}{3}}\right] = L\left[\frac{\frac{2a+b}{4}}{\frac{3a+b}{3}}\right]$$

$$\boxed{X_c = \frac{3L}{4}\left[\frac{2a+b}{3a+b}\right]}$$

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**QUESTION 2** (HW)

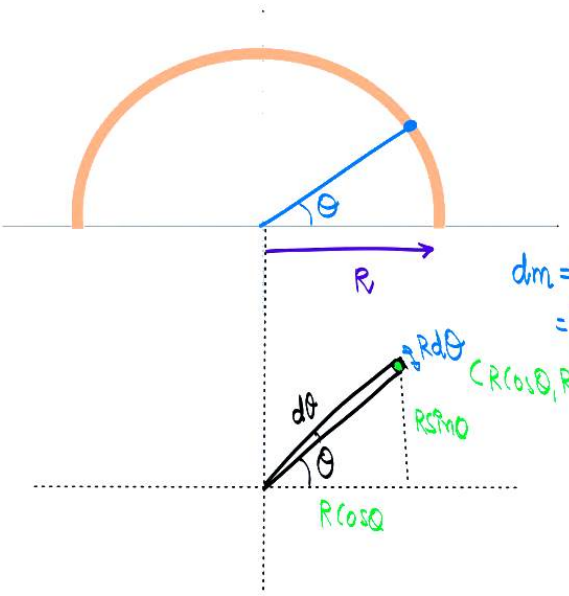
$\lambda = \lambda_0 \sin\theta$.

(Non uniform Semi Ring).

$Y_c$ → y coordinate of COM

$$Y_c = \frac{\int y\,dm}{\int dm}$$

y element




$dm = \lambda\,dx$

$= (\lambda_0 \sin\theta)R\,d\theta$

$$Y_c = \frac{\int (R\sin\theta)(\lambda_0 R\sin\theta\,d\theta)}{\int \lambda_0 R\sin\theta\,d\theta} = \frac{\cancel{\lambda_0}R^2\int_0^\pi \sin^2\theta\,d\theta}{\cancel{\lambda_0}R\int_0^\pi \sin\theta\,d\theta}$$

(100%)

$$= R\,\frac{\int_0^\pi \left[\frac{1-\cos 2\theta}{2}\right]d\theta}{[-\cos\theta]_0^\pi}$$

$$= \frac{R}{2}\,\frac{\left[\int_0^\pi d\theta - \int_0^\pi \cos 2\theta\,d\theta\right]}{[-\cos\theta]_0^\pi}$$

$$= \frac{\frac{R}{2}\left[[\theta]_0^{\pi} - \left[\frac{\sin 2\theta}{2}\right]_0^{\pi}\right]}{[-\cos\theta]_0^{\pi}}$$

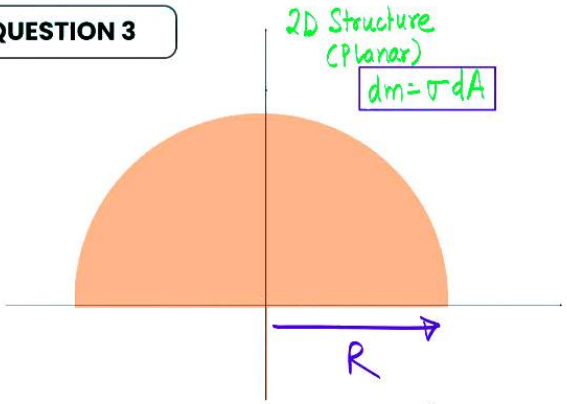
$$= \frac{\frac{R}{2}\left[\pi - \frac{1}{2}[\sin 2\pi - \sin 0]\right]}{-[\cos\pi - \cos 0]}$$

$$-[-1-1]$$

$$\boxed{y_c = \frac{\pi R}{4}} \text{ Ans.}$$

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**QUESTION 3**




Semi disc of Mass = M & Radius = R.

$Y_c = \frac{\int y\,dm}{\int dm}$ → y coordinate of COM of element

dm = mass of Element

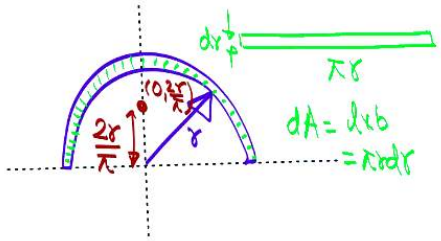
$\frac{\pi R^2}{2} \rightarrow M$

$1 \rightarrow \frac{M}{\frac{\pi R^2}{2}}$ (Areal Mass density)

$\pi r dr \rightarrow \frac{2M}{\pi R^2} \pi r dr$

$dm = \frac{2M r dr}{R^2}$

disc ⇒ Element = Ring




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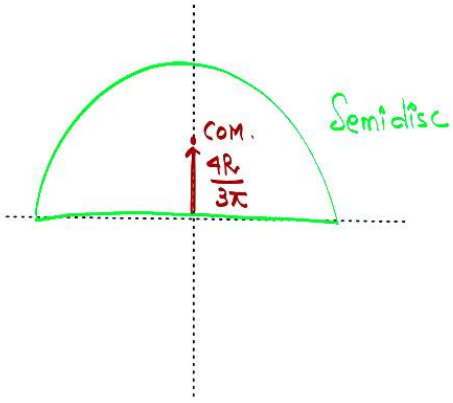
$$Y_c = \frac{\int_0^R \frac{2r}{\pi}\frac{2Mr\,dr}{R^2}}{\int_0^R \frac{2Mr\,dr}{R^2}} \quad \text{Variable}$$

$$= \frac{\frac{4M}{\pi R^2}\int_0^R r^2\,dr}{\frac{2M}{R^2}\int_0^R r\,dr}$$

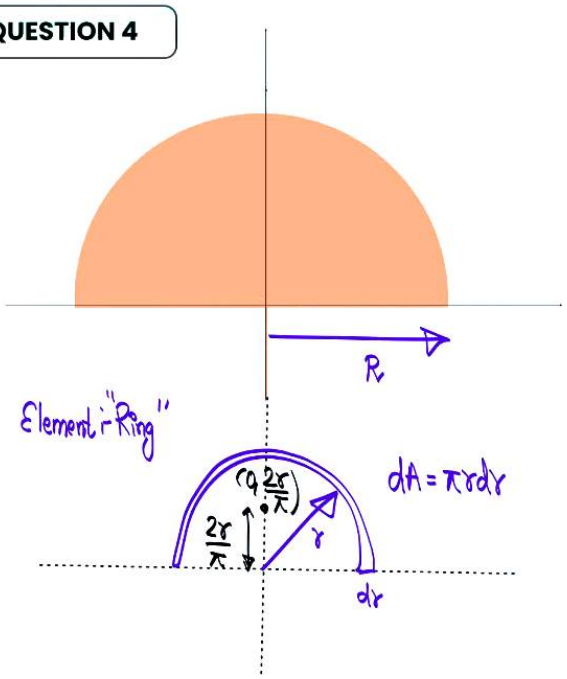
$$= \frac{2}{\pi}\,\frac{\left[\frac{r^3}{3}\right]_0^R}{\left[\frac{r^2}{2}\right]_0^R}$$

$$= \frac{2}{\pi} \times \frac{R^3}{3} \times \frac{2}{R^2}$$

$$= \frac{4R}{3\pi}$$



## QUESTION 4




Semi disc of Radius = R

has Non uniform Mass density.

$\sigma = \sigma_0 r^2$

where r is from origin.

$Y_c = \frac{\int y\,dm}{\int dm}$ → y coord of com of element

$dm = \sigma dA$

$= \sigma_0 r^2 \pi r dr$

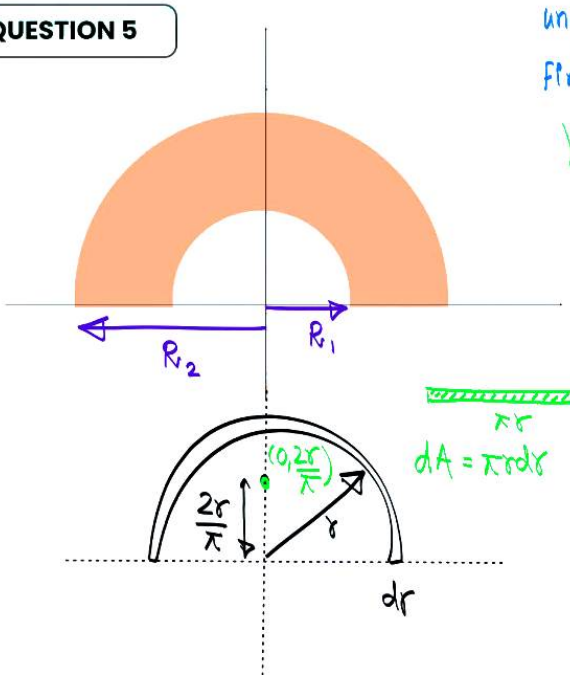
Variable $dm = \sigma_0 \pi r^3 dr$

$$= \frac{\int_0^R \frac{2r}{\pi} \times \sigma_0 \pi r^3 dr}{\int_0^R \sigma_0 \pi r^3 dr}$$

$$Y_c = \frac{\int_0^R \frac{2r}{\pi} \sigma_0 \pi r^3 dr}{\int_0^R \sigma_0 \pi r^3 dr} = \frac{2\pi\sigma_0 \int_0^R r^4 dr}{\sigma_0 \pi \int_0^R r^3 dr} = \frac{2}{\pi}\frac{R^5}{5} \times \frac{4}{R^4}$$

$$= \frac{8R}{5\pi}.$$

## QUESTION 5




uniform annular disc of Mass density = $\sigma$

Find $Y_{com} = ?$

$Y_{com} = \frac{\int y\,dm}{\int dm}$ → y coord of com of element.

$dm = \sigma dA$

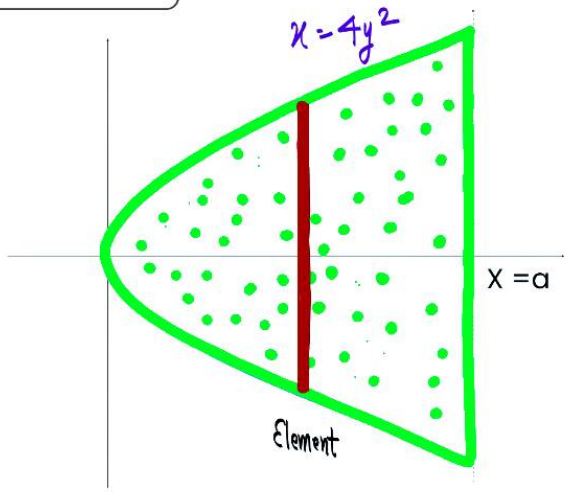
$= \sigma \pi r dr$

Variable.

$$= \frac{\int_{R_1}^{R_2} \frac{2r}{\pi} \cdot \sigma \pi r dr}{\int_{R_1}^{R_2} \sigma \pi r dr} = \frac{2\sigma \int_{R_1}^{R_2} r^2 dr}{\sigma\pi \int_{R_1}^{R_2} r dr}$$

$$Y_c = \frac{\frac{2}{\pi}\left[\frac{R_2^3}{3} - \frac{R_1^3}{3}\right]}{\left[\frac{R_2^2}{2} - \frac{R_1^2}{2}\right]}$$

**QUESTION 6**

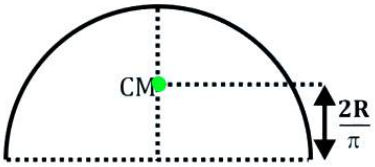



# Parabolic Sheet of uniform mass density $= \sigma$.

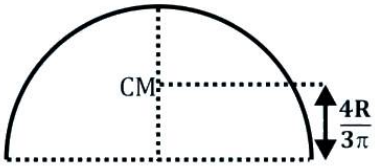
Find $X_c = ?$

❖ **Centre of mass of some uniform symmetric bodies are**

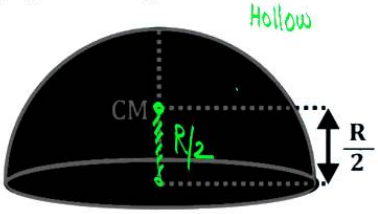
(i) Semicircular ring of radius R



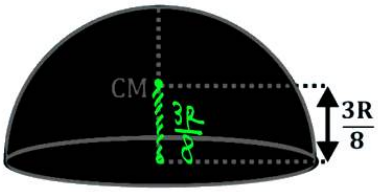

(ii) Semicircular disc



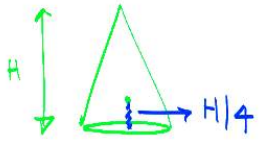

(iii) Hemispherical shell




(iv) Solid hemisphere

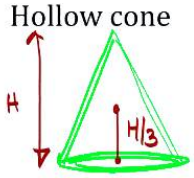


(v) Solid cone

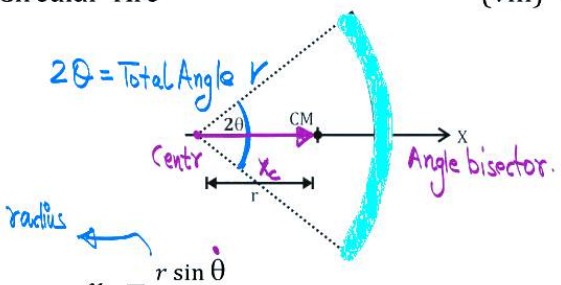



#

(vi) Hollow cone



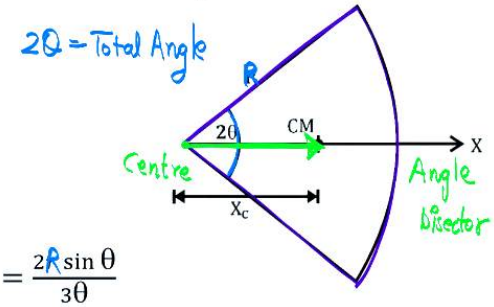

(vii) Circular Arc




$$x_c = \frac{r \sin\theta}{\theta}$$

Note: Here θ is in radians.

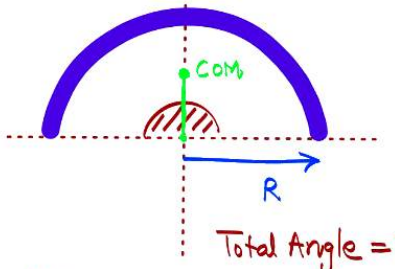
(viii) Sector of a circular plate




$$x_c = \frac{2R\sin\theta}{3\theta}$$

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Example.



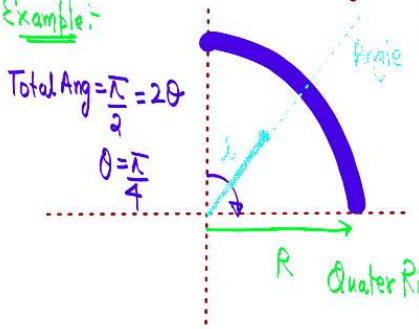

Total Angle = π = 2θ

θ = π/2

general $\frac{r\sin\theta}{\theta}$

$$y = \frac{R\sin(\pi/2)}{\pi/2} = \frac{2R}{\pi}$$

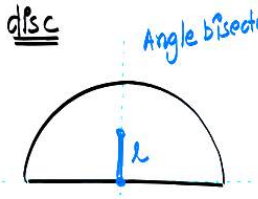
Example:-




$$l = \frac{R\sin(\pi/4)}{\pi/4} = \frac{4R}{\pi}\times\frac{1}{\sqrt{2}} = \frac{2\sqrt{2}R}{\pi}$$

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# disc

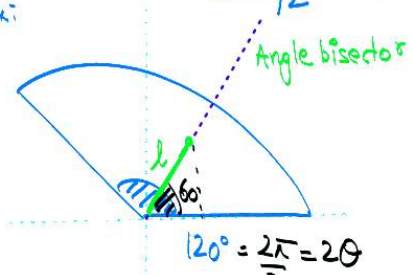



Semidisc $l = \frac{2R\sin\theta}{3\theta} = \frac{2R\sin(\pi/2)}{3(\pi/2)} = \frac{4R}{3\pi}$

Total Angle = π = 2θ

θ = π/2

Ex:




120° = 2π/3 = 2θ

θ = π/3

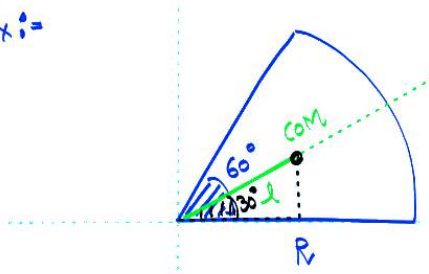
find Coordinate of COM :- = $l\cos 60, l\sin 60$

$= \frac{R\sqrt{3}}{\pi}\times\frac{1}{2}, \frac{R\sqrt{3}}{\pi}\times\frac{\sqrt{3}}{2}$

$= \left(\frac{R\sqrt{3}}{2\pi}, \frac{3R}{2\pi}\right)$

$l = \frac{2R\sin\pi/3}{3(\pi/3)} = \frac{R\sqrt{3}}{\pi}$

Ex :-




Coordinate of COM = $l\cos 30, l\sin 30$

$= \frac{2R\sqrt{3}}{\pi}\frac{1}{2}, \frac{2R}{\pi}\times\frac{1}{2}$

$= \left(\frac{R\sqrt{3}}{\pi}, \frac{R}{\pi}\right)$

Total Angle $= 2\theta = \frac{\pi}{3}$

$\theta = \frac{\pi}{6}$

disc

$l = \frac{2R\sin\theta}{3\theta}$

$= \frac{2R\sin\pi/6}{3(\pi/6)} = \frac{2R}{\pi}$

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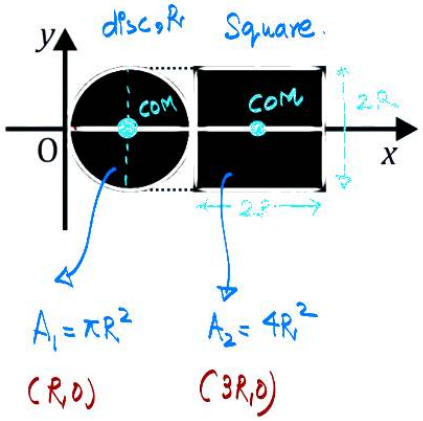
Additive | Negative System

- In terms of Mass
- 2D objects (In terms of Area)
- 3D (In terms of Volume)

| In terms of Mass | 2D objects (In terms of Area) | 3D (In terms of Volume) |
|---|---|---|
| Add $X_c = \frac{m_1x_1+m_2x_2+\cdots}{m_1+m_2+\cdots}$ | Add $X_c = \frac{A_1x_1+A_2x_2}{A_1+A_2}$ | Add $X_c = \frac{V_1x_1+V_2x_2}{V_1+V_2}$ |
| $Y_c = \frac{m_1y_1+m_2y_2+\cdots}{m_1+m_2+\cdots}$ | $Y_c = \frac{A_1y_1+A_2y_2}{A_1+A_2}$ | $Y_c = \frac{V_1y_1+V_2y_2}{V_1+V_2}$ |
| Neg ⇒ (Removal) $X_c = \frac{m_1x_1-m_2x_2-m_3x_3\cdots}{m_1-m_2-m_3-\cdots}$ | Neg $= X_c = \frac{A_1x_1-A_2x_2}{A_1-A_2}$ | Neg $X_c = \frac{V_1x_1-V_2x_2}{V_1-V_2}$ |
| $Y_c = \frac{m_1y_1-m_2y_2-m_3y_3\cdots}{m_1-m_2-m_3-\cdots}$ | $Y_c = \frac{A_1y_1-A_2y_2}{A_1-A_2}$ | $Y_c = \frac{V_1y_1-V_2y_2}{V_1-V_2}$ |

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**QUESTION 9** **Centre of mass of composite bodies:**




disc, R  Square

$A_1 = \pi R^2$ (R, 0)

$A_2 = 4R^2$ (3R, 0)

"Add" $X_c = \frac{A_1x_1+A_2x_2}{A_1+A_2}$

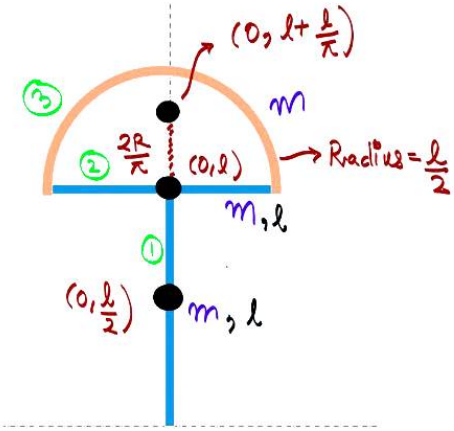
$= \frac{\pi R^2(R) + 4R^2(3R)}{\pi R^2+4R^2}$

$= \frac{\pi R^3+12R^3}{\pi R^2+4R^2} = R\left[\frac{12+\pi}{4+\pi}\right]$

$Y_c = 0$

## QUESTION 10

"Add"




$$Y_c = \frac{m_1y_1 + m_2y_2 + m_3y_3}{m_1+m_2+m_3}$$

$$= \frac{m\left(\frac{l}{2}\right) + m(l) + m\left(l+\frac{l}{\pi}\right)}{3m}$$

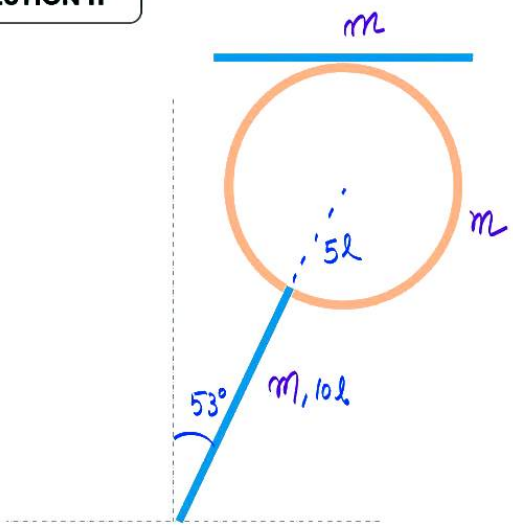
① m $(0, \frac{l}{2})$
② m $(0, l)$
③ m $(0, l+\frac{l}{\pi})$

$$= \frac{\frac{3ml}{2} + ml\left(1+\frac{1}{\pi}\right)}{3m}$$

$$Y_c = \frac{l}{2} + \frac{l}{3}\left(1+\frac{1}{\pi}\right)$$

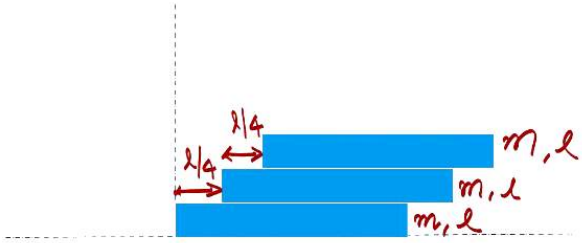
$X_c = 0$

## QUESTION 11



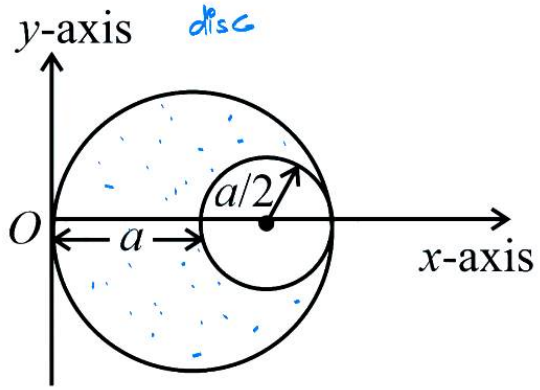

## QUESTION 12

3 Rods



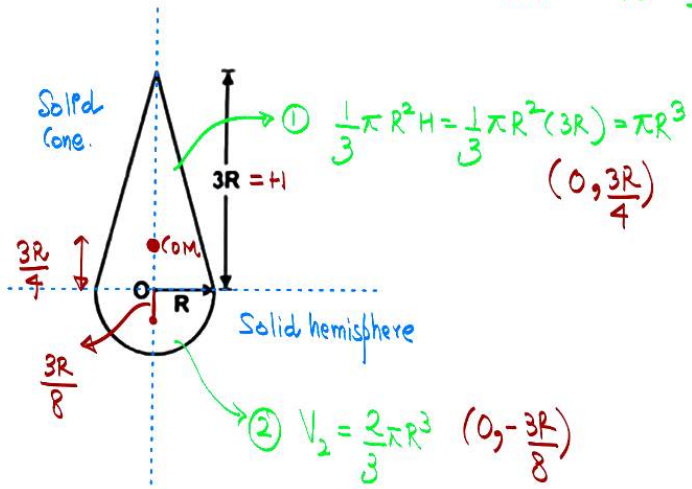

$X_c = ?$

**QUESTION 13**




**QUESTION 14**

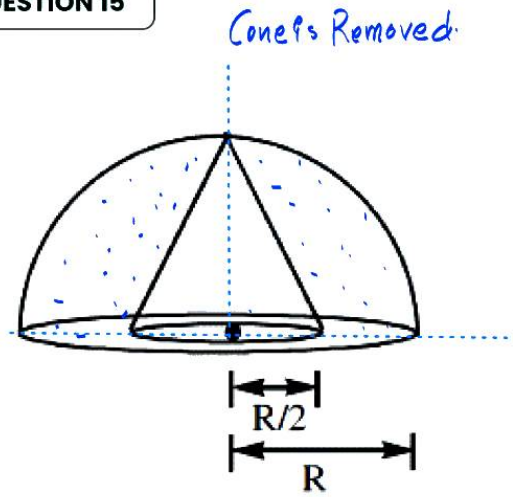
"Add" $Y_c = \frac{V_1 y_1 + V_2 y_2}{V_1 + V_2} = \frac{\pi R^3\left(\frac{3R}{4}\right) + \frac{2}{3}\pi R^3\left(-\frac{3R}{8}\right)}{\pi R^3 + \frac{2}{3}\pi R^3}$




① $\frac{1}{3}\pi R^2 H = \frac{1}{3}\pi R^2 (3R) = \pi R^3$ $\left(0, \frac{3R}{4}\right)$

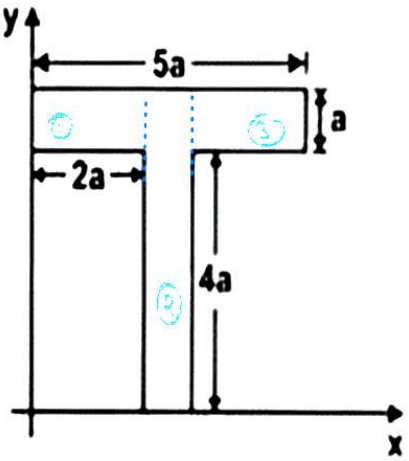
② $V_2 = \frac{2}{3}\pi R^3$ $\left(0, -\frac{3R}{8}\right)$

**QUESTION 15**



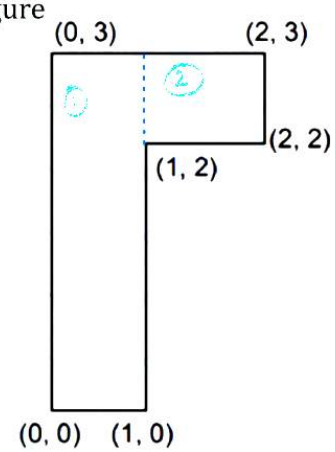
**QUESTION 16**

Figure shows a uniform two dimensional T-shaped structure. Find co-ordinates of centre of mass of this object.



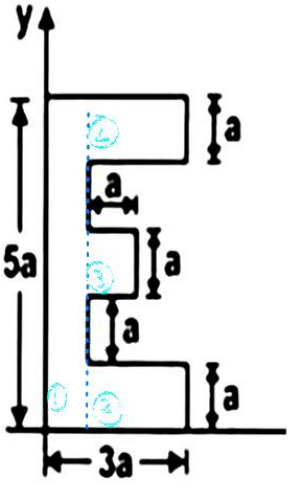

**QUESTION 17**

Find the coordinates of centre of mass of the lamina, shown in figure



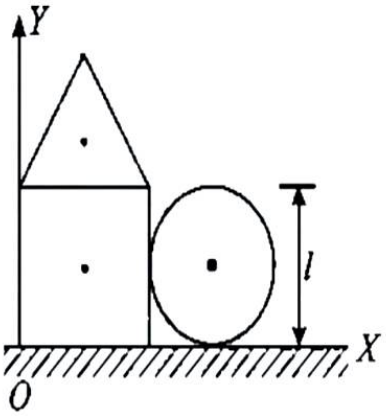

**QUESTION 18**

Figure shows a uniform plate oi shape *E* as shown. Find the co-ordinates of centre of mass of this structure.



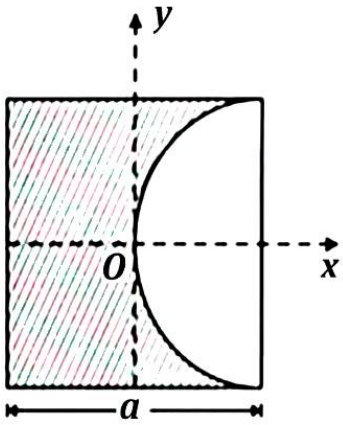
**QUESTION 19**

Three laminar objects of uniform density a square, a disc and an equilateral triangle are placed as shown in figure . Find the coordinates of center of mass of the three bodies.



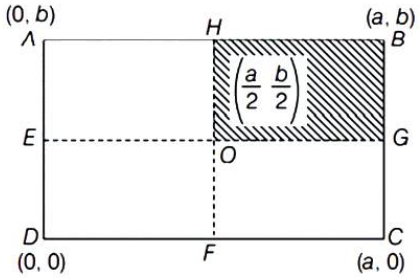

**QUESTION 20**

Figure shows a square plate of side $a$ from which a semicircular disc is removed as shown. Find the centre of mass of the remaining object.




**QUESTION 21**

A uniform rectangular thin sheet *ABCD* of mass M has length a and breadth b, as shown in the figure. If the shaded portion *HBGO* is cut-off, the coordinates of the centre of mass of the remaining portion will be




**A** $\left(\frac{2a}{3}, \frac{2b}{3}\right)$

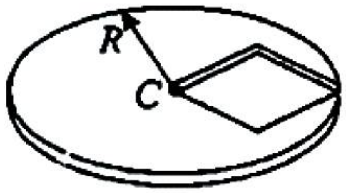
**B** $\left(\frac{5a}{12}, \frac{5b}{12}\right)$

**C** $\left(\frac{3a}{4}, \frac{3b}{4}\right)$

**D** $\left(\frac{5a}{3}, \frac{5b}{3}\right)$

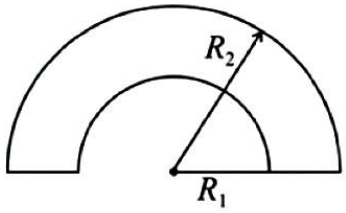
**QUESTION 22**

Find the centre of mass of the disc shown in figure. A square is removed from the disc as shown .



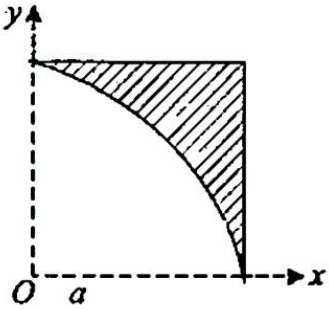

**QUESTION 23**

Find the centre of mass of an annular half disc shown in figure.



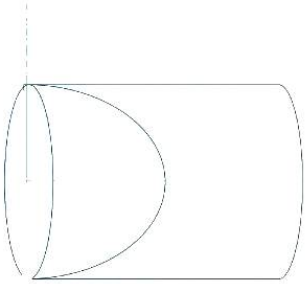

**QUESTION 24**

From a square plate of side $a$, $a$ quarter circular disc of radius $a$ is removed as shown in figure. Find the coordinates of center of mass of the remaining part.



**QUESTION 25**

From a cylinder of length 2R and radius R, a hemispherical portion is removed, find COM of remaining portion.



**QUESTION 26**

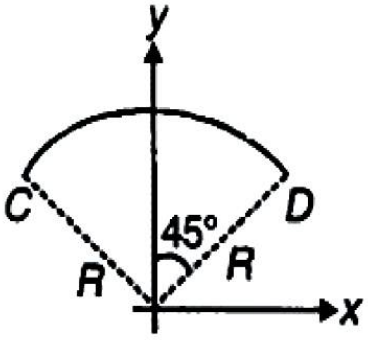
A circular arc $CD$ of thin wire frame of radius $R$ and mass $M$ with center at origin makes an angle of 90° at the origin. The $y$-coordinate center of mass of the arc lies at

**A** $\frac{\sqrt{2}R}{\pi}$

**B** $\frac{R}{2\pi}$

**C** $\frac{2R}{\pi}$

**D** $\frac{2\sqrt{2}R}{\pi}$






## Homework

- Complete Questions from module till topic of finding COM.

❖ Cover you Backlog or revise effectively from JEE Wallah channel:




Thank You